Science of the Total Environment 906 (2024) 167465

Contents lists available at ScienceDirect

Science of the Total Environment

journal homepage: www.elsevier.com/locate/scitotenv

Seagrass meadows as ocean acidification refugia for sea urchin larvae

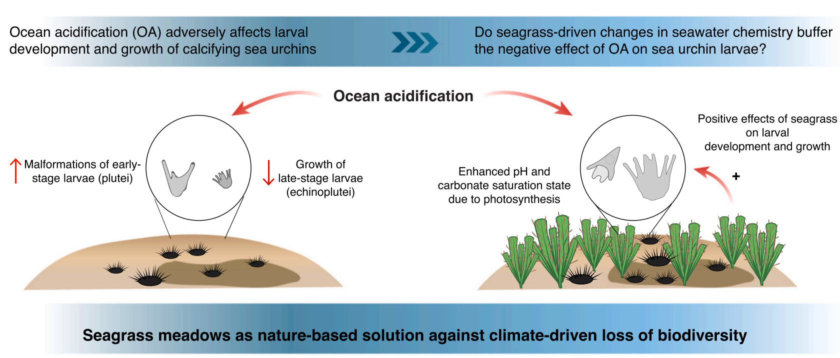
C. Ravaglioli[a,*,1], L. De Marchi[b,1], J. Giannessi[b], C. Pretti[b,c], F. Bulleri[a,d]

[a] Dipartimento di Biologia, Università di Pisa, CoNISMa, via Derna 1, 56126 Pisa, Italy
[b] Dipartimento di Scienze Veterinarie, Università of Pisa, Via Livornese (lato monte) - 56122 San Piero a Grado, Pisa, Italy
[c] Interuniversity Consortium of Marine Biology and Applied Ecology "G. Bacci" (CIBM), Viale N.Sauro 4, 57128 Livorno, Italy
[d] Centro interdipartimentale di Ricerca per lo Studio degli Effetti del Cambiamento Climatico (CIRSEC), Università di Pisa, Italy

HIGHLIGHTS

- Sea urchin larvae were grown at ambient or low pH in tanks with or without plants.
- Plant photosynthesis increased pH and altered seawater chemistry at both pH levels.
- Positive effects of plant metabolism on larval development and growth at low pH
- Seagrass meadows as a tool against climate-driven loss of calcifying species

GRAPHICAL ABSTRACT



ARTICLE INFO

Editor: Olga Pantos

Keywords:
Mitigation
Ocean acidification
pH fluctuation
Climatic refugia
Seagrass
Sea urchin larvae

ABSTRACT

Foundation species have been widely documented to provide suitable habitats for other species by ameliorating stressful environmental conditions. Nonetheless, their role in rescuing stress-sensitive species from adverse conditions due to climate change remains often unexplored. Here, we performed a mesocosm experiment to assess whether the seagrass, *Posidonia oceanica*, through its photosynthetic activity, could mitigate the negative effects of ocean acidification on larval development and growth of the calcifying sea urchin, *Paracentrotus lividus*. Sea urchin larvae at early and late developmental stages that are generally associated to benthic habitats, were grown in aquaria with or without *P. oceanica* plants, under ambient or low pH conditions predicted by the end of the century under the worst climate scenario (RCP8.5). The percentage of abnormal larvae and their total body length under different experimental conditions were assessed on early- (i.e., pluteus; 72 h post-fertilization) and final-developmental stages (i.e., echinopluteus; 30 days post-fertilization), respectively. The presence of *P. oceanica* increased mean daily pH values of ~0.1 and ~0.15 units at ambient and low pH conditions, respectively, compared with tanks without plants. When grown at low pH in association with *P. oceanica*, plutei showed a ~23 % reduction of malformations and echinoplutei a ~34 % increase in total body length, respectively, compared with larvae developing in tanks without plants. Our results suggest that *P. oceanica*, by increasing pH and altering seawater carbonate chemistry through its metabolic activity, could buffer the negative

* Corresponding author.
E-mail addresses: chiara.ravaglioli@biologia.unipi.it (C. Ravaglioli), lucia.demarchi@unipi.it (L. De Marchi), joanna.giannessi@phd.unipi.it (J. Giannessi), carlo.pretti@unipi.it (C. Pretti), fabio.bulleri@unipi.it (F. Bulleri).
[1] Chiara Ravaglioli and Lucia De Marchi contributed equally to the work.

https://doi.org/10.1016/j.scitotenv.2023.167465
Received 3 July 2023; Received in revised form 15 September 2023; Accepted 28 September 2023
Available online 29 September 2023
0048-9697/© 2023 The Authors. Published by Elsevier B.V. This is an open access article under the CC BY license (http://creativecommons.org/licenses/by/4.0/).

effects of ocean acidification on calcifying organisms and could, thus, represent a tool against climate-driven loss of biodiversity.

1. Introduction

Global climate changes, due to rising anthropogenic CO_2 emissions, are causing unprecedented alterations to marine ecosystems (Cornwall et al., 2021; Doney et al., 2009; Kroeker et al., 2013; Nagelkerken and Connell, 2015) that are generally the result of an interplay between physiological impairment and altered species interactions (Bulleri et al., 2018; Gilman et al., 2010; Kroeker et al., 2017; Sunday et al., 2017). Many coastal communities are organized by hierarchical interactions in which facilitation by foundation species, such as corals, salt marsh plants, mangroves, seagrasses and kelp, are of primary importance (Altieri et al., 2007; Gribben et al., 2009). Thus, understanding their role in regulating the response of associated species to climate changes is paramount to devise sound conservation and restoration strategies.

Anthropogenic ocean acidification (OA) has been widely demonstrated to negatively affect calcifying organisms, by impairing their capacity to maintain carbonate structures and altering acid-base regulation (Byrne and Hernández, 2020; Doney et al., 2009; Hall-Spencer et al., 2008; Kroeker et al., 2010; Kurihara, 2008). Sea urchins are considered to be particularly vulnerable to OA because they calcify in both their planktonic larval stage and benthic adult life (Byrne et al., 2013; Byrne and Hernández, 2020; Dworjanyn and Byrne, 2018). As shown for other calcifying taxa, their early development stages appear to be more sensitive to OA compared to adults, likely representing a major bottleneck for sea urchin populations under future climate scenarios (Byrne et al., 2013; Byrne and Hernández, 2020; Kurihara, 2008; Lamare et al., 2016). Previous laboratory studies reported reduced growth, delayed development and increased body abnormality of larvae exposed to constant low pH conditions (reviewed by Byrne and Hernández, 2020). These negative effects have been explained in terms of energetic constraints, impaired digestion, and altered metabolism, acid-base regulation and calcification during different life-history stages of larvae (Chan et al., 2011; Stumpp et al., 2013, 2012, 2011). On the other hand, sea urchin larvae inhabiting natural fluctuating pH environments, such as upwelling regions, intertidal pools and CO_2 vents, appear to be more tolerant to future OA, as a result of a complex interplay of biotic (e.g., parental acclimation, phenotypic plasticity) and abiotic (e.g., food availability) factors at play (Byrne et al., 2020; Foo et al., 2020; Karelitz et al., 2020; Migliaccio et al., 2019; Uthicke et al., 2016). Within this context, understanding how the responses of sensitive species to OA could be influenced by local environmental variability is key to identify climate refugia and devising sound mitigation strategies (Kapsenberg and Cyronak, 2019).

Coastal areas are generally characterized by diel fluctuations in seawater carbonate chemistry, mainly driven by metabolic activities of primary producers (Falkenberg et al., 2021; Torres et al., 2021). In particular, habitat-forming species, such as seagrasses and macroalgae, can generate local areas of elevated pH, with increases ranging from about 0.06 up to a full unit during the day (Hendriks et al., 2014; Pacella et al., 2018; Pfister et al., 2019; Ricart et al., 2021b). This effect is generally restricted to the environment surrounding macrophyte canopies (from the boundary layer to meters) and varies with macrophyte density, light and hydrodynamic conditions (Hendriks et al., 2014; James et al., 2020; Kapsenberg and Cyronak, 2019; Krause-Jensen et al., 2016; Noisette et al., 2022; Noisette and Hurd, 2018; Ricart et al., 2021a; Wahl et al., 2018). Hence, dense vegetated habitats could represent a local climatic refugia for associated calcifying organisms, by either enhancing their adaptive capacity to fluctuating environment or temporally reducing exposure to low pH (Falkenberg et al., 2021; Noisette and Hurd, 2018; Ricart et al., 2021a; Wahl et al., 2018).

To date, the evaluation of the role of vegetated habitats as OA refugia has generated contrasting results across calcifying taxa and life-history stages (Kapsenberg and Cyronak, 2019). Previous laboratory studies on adult bivalves showed that macroalgae, by increasing the overall mean pH through their photosynthetic activity, promoted the calcification and growth of calcifying organisms under OA scenario (Wahl et al., 2018; Young et al., 2022; Young and Gobler, 2018). In the field, due to low macroalgal biomass to water volume ratio, the impact of biogenic fluctuations on calcifies was less pronounced and restricted to the proximity of macroalgal stands (Wahl et al., 2018; Young et al., 2022). By contrast, a few recent studies on bivalve larvae reported inconsistent outcomes, depending on species and biological processes investigated and the timing of exposure to unfavourable conditions (Clark and Gobler, 2016; Frieder et al., 2014; Kapsenberg et al., 2018). For instance, Frieder et al. (2014) showed that semidiurnal fluctuations in pH mitigated the negative effects of low pH on the larval development of two mussel species, while pH variations enhanced the larval shell growth in *Mytilus galloprovincialis*, but not in *Mytilus californianus* under OA. In contrast, Kapsenberg et al. (2018) found negative effects of low pH on the larval growth of *M. galloprovincialis*, regardless of pH variations, and an increase in the larval abnormal development due to the exposure to low pH during specific soft-tissue development stages. These investigations, however, were performed under simulated (chemically) pH fluctuations and in the absence of species interactions and, thus, it remains unclear how realistic and gradual pH variations generated by macrophytes could influence different larval development stages of vulnerable species under future climates.

Here, we performed a mesocosm study to assess whether the seagrass, *Posidonia oceanica*, through its metabolic activity, could mitigate the negative effects of OA on the development and growth of larval stages of the calcifying sea urchin, *Paracentrotus lividus*. In the field, the metabolism of *P. oceanica* during daytime has been shown to generate pH increases ranging from 0.06 (in September) to 0.24 (in June) pH units, both within and at the edge of the meadows, with the largest increases occurring at the peak in seagrass productivity and leaf area (Hendriks et al., 2014). Although sea urchins, including *P. lividus*, generally disperse as planktonic larvae for several days (García et al., 2015), the early and final development phases (i.e., shortly after gamete fertilization and before settlement) are likely to be associated with surrounding benthic habitats (Prado et al., 2009; Simon and Levitan, 2011; Steller and Cáceres-Martínez, 2009). In particular, specimens of *P. lividus* are commonly found within *P. oceanica* meadows (Ceccherelli et al., 2009), which provide greater availability of food and shelter than rocky shores and a suitable substrate for larval settlement (Pinna et al., 2012; Prado et al., 2009; Privitera et al., 2011). Thus, in this study, sea urchin larvae during different life-history stages (i.e., after fertilization and before settlement) were exposed to ambient and low pH conditions, in tanks with or without *P. oceanica*. The percentage of abnormal larval development and total body length were assessed on the early (72 h post-fertilization) and the late (30 days post-fertilization) development of larvae, respectively, under different experimental conditions. Specifically, we hypothesize that i) plants of *P. oceanica* would increase seawater pH and modify carbonate chemistry, through their photosynthetic activity (Hendriks et al., 2014), at both ambient and low pH conditions and ii) plant conditioning of pH condition would increase larval development and growth under OA.

2. Materials and methods

2.1. Sea urchin and plant collection and preparation

The experiment was run twice over two spawning periods of the sea urchin, *Paracentrotus lividus*, in June and November 2021 (the two main reproductive peaks of this species generally are in early Spring and in Autumn, Guettaf et al., 2000; Hereu et al., 2004). Individuals of *P. lividus* were collected at depths of ~2–3 m south of Livorno (Antignano, Italy, 43°29′17.39″N, 10°19′33.17″E), in late May and mid-November 2021. The urchins were rapidly transferred at the aquarium of Livorno and kept into a large holding tank with filtered seawater until the start of the experiment. Large rhizome fragments of *Posidonia oceanica* were also collected by divers from well-preserved meadows at a depth of ~5 m, located off the coast of Livorno (43° 28′ 45.005″ N, 10° 17′ 30.001″ E). Plant materials were rapidly transported into coolers to the mesocosm facility located at the aquarium of Livorno.

The mesocosm facility consists of four large (500 L) and independent glass aquaria, with individual water circulating, illumination and filtration system. Aquaria were placed in a temperature-controlled room, without other external light sources and filled in at the start of the experiment with filtered and UV sterilized seawater. Since the number of aquaria was not sufficient to properly replicate the experiment, we performed two independent experimental runs, as previously reported in other field and mesocosm studies (Clements et al., 2018; Lamare et al., 2016). For each run, sixteen plant fragments of similar size and vertical shoot number were carefully selected and individually attached to the bottom of four plastic trays (48 cm × 30 cm × 12 cm) filled with small cobbles (4 fragments per tray with 40 ± 3 and 38 ± 1 shoots tray^{-1} for the first and the second run, respectively). Two trays were then randomly placed in two aquaria, while other two trays containing only small cobbles were allocated in the other two aquaria without *P. oceanica* plants. Each aquarium was then assigned to each of the four combinations of pH (ambient vs. low pH) and *P. oceanica* plants (+P vs. —P) (Fig. S1 in Supplementary Materials). Each aquarium with and without *P. oceanica* plants was lit by three (two Silvermoon Reef Blu 895 and one Silvermoon Marine 895) and two (one Silvermoon Reef Blu 895 and one Silvermoon Marine 895) LED lamps, respectively, which allow the simulation of light diel fluctuation. Irradiance levels in the experimental aquaria were adjusted to mimic a shallow-water light intensity with a peak irradiance of ~300 μmol photons m^{-2} s^{-1} above the canopy (Dattolo et al., 2014; Ruocco et al., 2021). The irradiance level measured at about 20 cm from the bottom of the aquaria was ~100 μmol photons m^{-2} s^{-1} both within the canopy and in tanks without plants, according to observed field values within shallow *P. oceanica* canopies. Seawater temperature was independently controlled in each aquarium by a chiller (Teco TK 500) and maintained at a constant temperature of ~19.5 ± 0.12 °C and ~17.4 ± 0.016 °C for the first and the second experimental run, respectively, in line with temperatures recorded in the field at the time of plant and sea urchin collection.

2.2. *Experimental CO_2 manipulation*

Plants were allowed to acclimatize to laboratory conditions for ~ten days before the start of experiments. CO_2 was then bubbled into the sumps of aquaria assigned to low pH, controlled by an automatic CO_2 injection and pH-controlled feedback system. The pH levels in the aquaria were monitored with pH electrodes, which were allocated in the recirculating water system of each aquarium as a control for CO_2 addition. The electrodes provided feedback to a control system that regulate the pH levels in each experimental aquarium by bubbling CO_2 gas into the sumps as required. The low pH treatment simulated values expected by the end of the century under the RCP 8.5 scenario (Riahi et al., 2011), with a decrease in the mean pH of ~0.4 units. In particular, in the aquarium without *P. oceanica*, pH was held almost constant (between 7.7 and 7.8 pH) while, in the aquarium with *P. oceanica*, pH values were allowed to range between 7.7 and 7.9 to maintain the desired mean pH (~7.8) and incorporate pH variations due to plant metabolism.

During the first experimental run (June), pH_{NBS} was monitored within each tank several times a day between 10:00 and 18:00 h throughout the experiment, using a portable pH meter (Hach HQ2100). Salinity was measured weekly using a conductivity meter and total alkalinity samples were collected weekly from each aquarium and measured using an automated potentiometric titration with a Methrom 848 Titrino plus system and applying the Gran method (Sass and Ben-Yaakov, 1977). For the second experimental run (November), pH_{NBS} was continuously monitored through HOBO pH and temperature data loggers (Onset MX2501) along the course of the experiment. Since this experimental run lasted only three days, salinity was measured twice and total alkalinity samples were collected once during the experiment. Carbonate system variables were then calculated from measured pH, total alkalinity, temperature and salinity values using CO2SYS program for Excel with constant from Mehrbach et al. (1973) and adjusted by Dickson and Millero (1987) (see Tables S1 and S2 for the carbonate seawater chemistry during the early and the late larval development, respectively).

2.3. *Spawning and fertilization procedure*

Mature adults of *P. lividus* were induced to spawn by intracoelomic injection of 0.5 mL of 0.5 M KCl *per* animal (Espinel-Velasco et al., 2020). Eggs were collected into beakers of 100 mL with 0.45 μm filtered natural seawater. Sperms were stored in 1.5 mL Eppendorf tubes and conserved at 4 °C. Gametes were visually checked for quality in term of sperm motility and egg shape and colour. Spawned eggs were washed with filtered seawater, pooled into a 1 L beaker and, then, fertilized by using a sperm concentration of 10^5 sperm Lm^{-1}. Fertilization success was determined after 2 h as the proportion of eggs that had a fertilization envelope or exhibited cleavage (Espinel-Velasco et al., 2020; Martin et al., 2011).

2.4. *Early larval development under different experimental conditions (days post-fertilization, dpf = 0–3)*

To assess the effects of experimental treatments from the onset of gastrulation up to the early development of pluteus stage (48 h–72 h) larvae, after fertilization (dpf = 0) embryos were transferred into 50 mL modified falcon tubes (Sartori et al., 2016) and either suspended within *P. oceanica* canopies at about 20 cm from the bottom or maintained at the same height in aquaria without plants, for 3 days (Fig. S1). The falcon tubes were closed with 100 μm mesh caps to restrict loss of embryos but allowing maximum water exchange (Espinel-Velasco et al., 2020; Lamare et al., 2016). Six falcon tubes were randomly allocated to each tank. At the end of the experiment (dpf = 3), tubes were retrieved from the tanks and transported to the laboratory. Samples of larvae (50 mL) were fixed with buffered 4 % formaldehyde for examination by optical microscopy. One hundred larvae were counted for each replicate tube and developmental abnormalities recorded (Sartori et al., 2016). In particular, larvae at pluteus stage, pyramid-shaped and with fully developed arms were considered normal, while abnormal development at the stages of blastula, gastrula or plutei were considered malformed larvae (Fig. 1a, b).

2.5. *Larval growth at late development stage under different experimental conditions (dpf = 20–30)*

The effects of experimental treatments on the overall length (apical end + post-oral arm) of echinoplutei (the final development stage before settlement, dpf = 30) were assessed only in the first run (June), since most of the larvae were lost before their development to the echinopluteus phase in the second run (November). Larvae were kept in 3 L aquaria at a density of 2–5 individuals per mL (Espinel-Velasco et al., 2020) with 0.45 μm filtered seawater and maintained at a constant temperature of ~18 °C for ~3 weeks until reaching competency at about 20 days post-fertilization. During this period, seawater was changed every 2–3 days by siphoning out ~90 % of the water in the aquaria and larvae were fed daily with a mix of *Isochrysis galbana*, *Dunaliella*

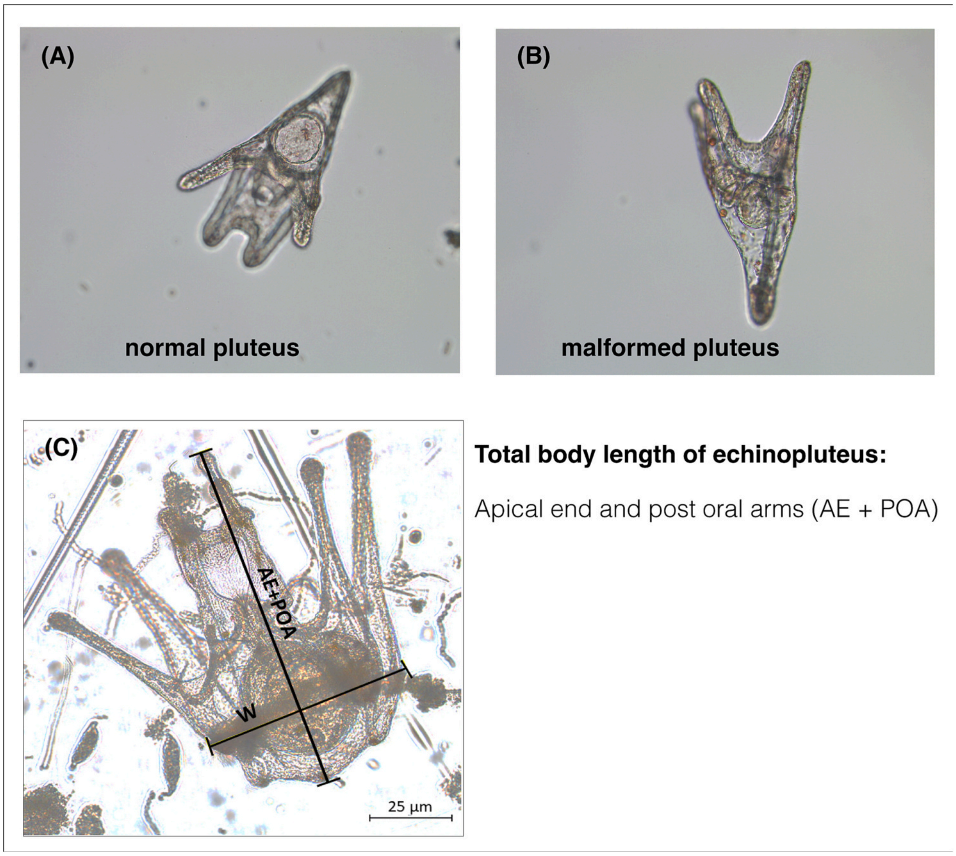


Fig. 1. Examples of (A) a normal larva with fully developed arms and (B) a malformed larva at pluteus stage at 3 days post-fertilization; (C) total body length of an echinopluteus larva measured as the total length of apical end plus post oral arms at 30 days post-fertilization.

tertiolecta and *Navicula* sp. (Culture Collection of Algae and Protozoa – Scottish Association for Marine Science/SAMS Research Services Ltd) at a concentration of approximately 7000–8000 cells *per* mL. After 20 days, fifty larvae were transferred into each falcon tube and randomly allocated to the four experimental aquaria (5 falcon tubes for each tank), as described above (Fig. S1). Larvae were maintained under different experimental conditions for ten days. At the end of the experiment (dpf = 30), tubes were retrieved from the tanks, transported to the laboratory and fixed in paraformaldehyde as described above. All larvae were photographed using a Leica DMi1 microscope fitted with an MC120-HD digital camera. The total arm length of larvae was then measured using the free software Image J (Fig. 1c).

2.6. Statistical analyses

Generalized linear mixed models (GLMMs), assuming a Gaussian distribution and an identity link, were used to analyse the effects of experimental treatments on both physical (mean daily pH values during early and late larval development phases) and biological (the abnormal development and the total body length of early and late larval stages, respectively) response variables. A GLMM was used to test whether the mean daily pH during the early larval development (*dpf* = 0–3) differed between *P. oceanica* (+P vs. —P) and pH (ambient vs. low) treatments over the two experimental runs (run1 vs. run2). The factors *P. oceanica*, pH and run were included in the fixed part of the model as predictor variables, while the factor tank was added as a random effect to take into account that replicate pH measurements were taken within each tank. For the mean daily pH during the late larval development (*dpf* = 20–30), the factors pH and *P. oceanica* were included in the fixed part of the model as predictor variables, and tank as a random effect. The percentage of pluteus larvae showing an abnormal development (*dpf* = 0–3) was analysed using a GLMM, including the factors pH and *P. oceanica* in the fixed part of the model as predictor variables. The factor tank nested in experimental run (run1 vs. run2) was added in the random part of the model to account for variation among tanks and experimental runs. For the total body length of echinopluteus larvae (*dpf* = 20–30), the model included pH and *P. oceanica* as fixed factors and tank as a random effect to take into account that replicated tubes were grouped within each tank. The GLMMs were run using *glmmTMB* R package. Post-hoc comparisons were performed with R function *emmeans* for the significant interaction terms. Residuals were checked with Q-Q plots and plots of standardized residuals versus expected values, using *DHARMA* R package. The same package was used to run a Kolmorov-Smirnoff test to formally assess heteroscedasticity and goodness-of-fit tests on simulated residuals to check for over-dispersion and outliers (Figs. S2–S5 in Supplementary Materials). All statistical analyses were performed in R version 3.6.1.

3. Results

3.1. Seawater carbonate chemistry

During the early larval development (*dpf* = 0–3), the mean daily pH was significantly higher at ambient than low pH treatments ($Estimate_{(pH)} = -0.52 \pm 0.03$, $p < 2e^{-16}$; Table S3, Fig. 2) and in aquaria with *P. oceanica* than those without plants ($Estimate_{(P.oceanica)} = 0.09 \pm 0.03$, $p = 0.003$, Table S3, Fig. 2), at both experimental runs. In addition, at ambient pH, the mean pH was higher during the first than the second run, while there were no differences in the mean pH between experimental runs at low pH ($Estimate_{(run\ x\ pH)} = 0.14 \pm 0.04$, $p = 0.001$; Table S3, Fig. 2). In particular, during the first experimental run (spring), mean daily pH (±SE) in ambient pH aquaria was 8.32 ± 0.02

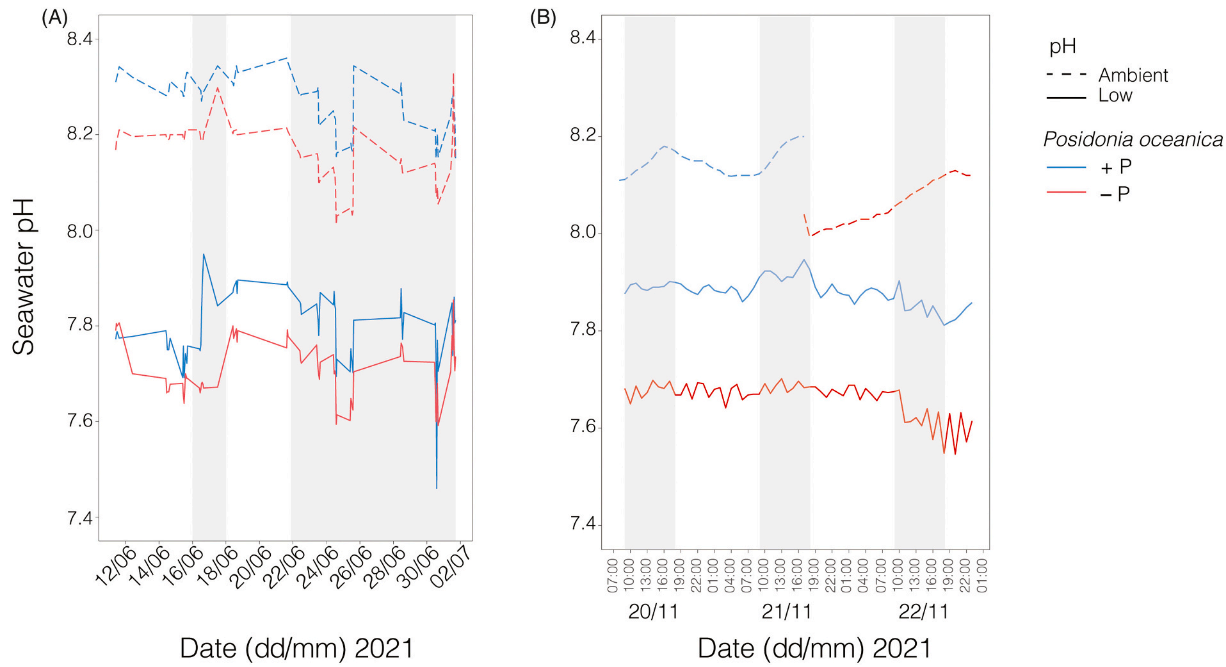


Fig. 2. (A) Time series of daily seawater pH values, measured between 10:00 and 18:00 throughout the experiment, for different combinations of pH (ambient and low) and *Posidonia oceanica* (+P and —P). Grey boxes indicate the early (dpf = 0–3) and the late larval development (dpf = 20–30) during the first run (June 2021). (B) Time series of hourly seawater pH for different combinations of pH (ambient and low) and *Posidonia oceanica* (+P and —P), measured during the second experimental run (November 2021). Grey boxes indicated pH values measured between 10:00 and 18:00 h for each day. For the second run, pH measurements were obtained by using three high-resolution pH HOBO data loggers, two of which were deployed in the two tanks exposed to low pH treatments (solid lines) throughout the experiment, while the third sensor was alternated between tanks with and without *P. oceanica* plants maintained at ambient pH (dotted lines).

(pCO_2 = 315.5 ± 14.2 μatm, Ω_{ca} = 6.60 ± 0.16 and Ω_{ar} = 4.29 ± 10) and 8.23 ± 0.03 (pCO_2 = 438.2 ± 0.8 μatm, Ω_{ca} = 5.68 ± 0.05 and Ω_{ar} = 3.69 ± 0.03) with and without *P. oceanica*, respectively. In low pH aquaria, mean pH was 7.85 ± 0.01 (pCO_2 = 1083 ± 64.8 μatm, Ω_{ca} = 2.86 ± 0.17 and Ω_{ar} = 1.86 ± 0.11) and 7.71 ± 0.04 (pCO_2 = 1554 ± 204 μatm, Ω_{ca} = 2.31 ± 0.31 and Ω_{ar} = 1.51 ± 0.20) with and without *P. oceanica*, respectively (see detailed results in Table S1, Fig. 2a). During the second experimental run (fall), mean pH (± SE) in ambient pH aquaria was 8.16 ± 0.01 (pCO_2 = 463.8 μatm, Ω_{ca} = 4.82 and Ω_{ar} = 3.12) and 8.05 ± 0.04 (pCO_2 = 649 μatm, Ω_{ca} = 4.10 and Ω_{ar} = 2.65) with and without *P. oceanica*, respectively. In low pH aquaria, mean pH was 7.88 ± 0.01 (pCO_2 = 969.2 μatm, Ω_{ca} = 2.91 and Ω_{ar} = 1.89) and 7.66 ± 0.02 (pCO_2 = 1807.1 μatm, Ω_{ca} = 1.91 and Ω_{ar} = 1.23) with and without *P. oceanica*, respectively (Table S1, Fig. 2b).

During the late larval development (*dpf* = 20–30) in June, the mean daily pH was significantly higher at ambient than low pH levels ($Estimate_{(pH)} = -0.42 \pm 0.02$, $p < 2e^{-16}$; Table S4, Fig. 2a) and in aquaria with *P. oceanica* than those without plants ($Estimate_{(P.oceanica)} = 0.11 \pm 0.02$, $p = 2.81e^{-07}$; Table S4, Fig. 2a). In particular, mean pH (±SE) in ambient pH aquaria was 8.24 ± 0.01 (pCO_2 = 327.4 ± 8.67 μatm, Ω_{ca} = 6.42 ± 1.12 and Ω_{ar} = 4.18 ± 0.07) and 8.12 ± 0.02 (pCO_2 = 495.5. ± 11.15 μatm, Ω_{ca} = 5.29 ± 0.12 and Ω_{ar} = 3.45 ± 0.07) with and without *P. oceanica*, respectively. In low pH aquaria, mean pH was 7.79 ± 0.02 (pCO_2 = 1078.8 ± 37.68 μatm, Ω_{ca} = 2.91 ± 0.11 and Ω_{ar} = 1.90 ± 0.07) and 7.70 ± 0.02 (pCO_2 = 1447.4 ± 51.77 μatm, Ω_{ca} = 2.45 ± 0.1 and Ω_{ar} = 1.59 ± 0.06) with and without *P. oceanica*, respectively (Table S2, Fig. 2a).

3.2. Larval development and growth under experimental conditions

There was a significant interaction between pH and *P. oceanica* on the percentage of abnormal development of pluteus larvae at dpf 3 (Table 1, Fig. 3). At low pH, larvae growing within *P. oceanica* canopies displayed a significant reduction (~23 %) in the percentage of abnormal development compared to those growing without plants while, at ambient pH, there were no differences in the percentage of abnormal larvae between *P. oceanica* treatments (Fig. 3).

Table 1
Results of a Generalized Linear Mixed Model (GLMM) used to assess the effects of pH (ambient = ApH, low = LpH) and *P. oceanica* (+P, —P) on the percentage of abnormal development of pluteus larvae at 3 days post-fertilization. Coefficients and standard errors (SE) for pH and *P. oceanica* are reported for the fixed effects, while estimates of variance (σ^2) and standard deviation (SD) per tank nested in run are reported for the random effects. ***P* < 0.01, ****P* < 0.001.

Effect	Abnormal development
Fixed effects	Estimate (SE)
Intercept	50.66 (3.57)***
+P	-2.83 (3.14)
LpH	26.33 (3.14)***
+P x LpH	-13 (4.45)**
Random effects	σ^2 (SD)
Tank (run)	3.1e-11 (5.6e-06)
Run	1.6e+01 (3.9)
Residual	5.9e+01 (7.7)
	Post-hoc contrasts
	(pH x P. oceanica)
	ApH: +P = -P
	LpH: +P < -P***

There was also a significant interaction between pH and *P. oceanica* on the total body length of echinopluteus larvae at dpf 30 (Table 2, Fig. 4). At low pH, the total body length of larvae growing within *P. oceanica* plants was significant higher (~34 %) compared to those maintained without plants while, at ambient pH, there were no differences in the larval body length between *P. oceanica* treatments (Table 2, Fig. 4).

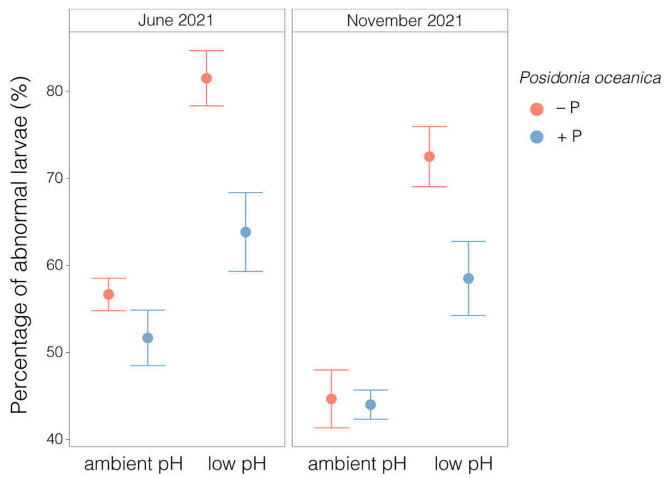


Fig. 3. Percentage of abnormal development of pluteus larvae (mean ± SE, *n* = 6) growing with or without *P. oceanica* plants under ambient and low pH conditions, for both experimental runs.

Table 2
Results of a Generalized Linear Mixed Model (GLMM) used to assess the effects of pH (ambient = ApH, low = LpH) and *P. oceanica* (+P, —P) on the total body length of echinopluteus larvae at 30 days post-fertilization. Coefficients and standard errors (SE) for pH and *P. oceanica* are reported for the fixed effects, while estimates of variance (σ^2) and standard deviation (SD) per tank are reported for the random effects. *$P < 0.05$, **$P < 0.01$, ***$P < 0.001$.

Effect	Total body length
Fixed effects	Estimate (SE)
Intercetta	92.5 (3.9)***
+P	2.7 (5.6)
LpH	−29.7 (5.6)***
+P x LpH	33.3 (7.9)***
Random effects	σ^2 (SD)
Tank	6.1e-08 (0.0002)
Residual	8e+01 (8.932)
	Post-hoc contrasts (pH x *P. oceanica*)
	ApH: +P = -P
	LpH: +P > -P

4. Discussion

Posidonia oceanica, by increasing the mean daily pH through its photosynthetic activity, mitigated the negative impacts of ocean acidification on the larval development and total body length of the sea urchin, *Paracentrotus lividus*, at key developmental stages. At low pH, early-stage larvae growing within *P. oceanica* canopies displayed a decrease of ~23 % of malformations. Likewise, the association with *P. oceanica* caused an increment of ~34 % of the total body length of echinoplutei during the final development stages that precede the settlement.

Ocean acidification has been widely shown to adversely affect different life-cycle stages of sea urchins (Byrne et al., 2013; Byrne and Hernández, 2020; Lamare et al., 2016), by altering physiological processes due to hypercapnia and impairing calcification (reviewed by Byrne and Hernández, 2020). In particular, sea urchin larvae seem to be the most vulnerable to OA (Byrne et al., 2013; Lamare et al., 2016; Stumpp et al., 2012), thereby representing a key life-history stage in determining future population dynamics and distributions. In our study, larvae growing at low pH without plants displayed a significant increase in the percentage of abnormal larvae during the first three days of development. This result is in line with previous reports of altered body morphology, including abnormal development and larval asymmetry, for a wide range of sea urchin larvae grown at low pH (reviewed by Byrne et al., 2013; Byrne and Hernández, 2020). Malformation of larvae growing at low pH has been suggested to be the result of energetic constraints associated to the alteration of metabolism, disruption of extracellular acid-base regulation and impaired digestion and calcification during embryonic/early larval development stages (Byrne et al., 2013; Byrne and Hernández, 2020; Foo et al., 2020; Stumpp et al., 2012, 2011).

At low pH and in the absence of *P. oceanica*, there was also a significant decrease in the total body length of larvae at the developmental stage at which they achieve competence for settlement (i.e., echinoplutei; ~one month post-fertilization). The presence of smaller echinoplutei under low pH condition may be a consequence of delayed larval development across life history stages (Byrne and Hernández, 2020; Stumpp et al., 2011) or reduced production of their biomineral mass (Byrne et al., 2013). In our study, larvae were grown under ambient conditions for ~three weeks and, then, exposed to low pH for the last ten days preceding settlement. Thus, at low pH, reduced growth of echinoplutei during their final planktonic stage could be due to hypercapnia, with a consequent reduction in skeleton production and calcification, rather than to the alteration of developmental timing (Byrne et al., 2013; Stumpp et al., 2012). Since arm length and larval skeleton are positively correlated to feeding success and swimming ability of larvae (Byrne and Hernández, 2020), the negative impacts of OA on larval size and body morphology during different development phases are likely to reduce their performance, thus compromising success of pelagic life stage and settlement.

The effects of OA on stress-sensitive species can be, however, influenced by local environmental conditions upon which climate change manifests (Falkenberg et al., 2021; Kapsenberg and Cyronak, 2019). Recent studies have shown that calcifying organisms (mainly sea urchins), inhabiting fluctuating pH environments, such as upwelling regions, tidal pools and CO_2 vents, enhanced their physiological tolerance to OA across different life stages, as a result of phenotypic plasticity and/or parental acclimation (Byrne et al., 2020; Kapsenberg and Cyronak, 2019; Karelitz et al., 2020; Migliaccio et al., 2019). Therefore, exposure to seawater chemistry variability has the potential to modulate sensitivity of calcifying species to OA and enhance their adaptive capacity (Kapsenberg and Cyronak, 2019). In many coastal areas, vegetated habitats, such as seagrass meadows and macroalgal forests, can modify the intensity of OA exposure for resident calcifying organisms, through their metabolic activities (i.e., photosynthesis and respiration), which can remove and add seawater CO_2 (Falkenberg et al., 2021; Pfister et al., 2019; Ricart et al., 2021b). For example, the photosynthetic activity of seagrasses, including *P. oceanica*, has been reported to enhance pH levels with a mean increase ranging from 0.07 to 0.15 units across different spatial and temporal scales compared to non-vegetated habitats (Hendriks et al., 2014; Pacella et al., 2018; Ricart et al., 2021b). Amelioration of low pH was generally more prolonged and of great intensity than events of reduced pH (Hendriks et al., 2014; Ricart et al., 2021b). This change in seawater chemistry could act as OA refugia for stress-sensitive species, by temporally reducing exposure to harmful conditions, although this benefit may vary with seasons, hydrodynamic conditions and macrophyte density and biomass (Falkenberg et al., 2021; Krause-Jensen et al., 2016; Ricart et al., 2021b; Wahl et al., 2018).

To date, the biological effects of pH variability on calcifying marine invertebrates remain somewhat elusive, with non-generalizable responses among taxa and life stages (Kapsenberg and Cyronak, 2019). For example, macrophyte-driven biogenic pH fluctuations have been shown to increase calcification and growth of adult bivalves under OA scenario, by shifting most of their calcification activity during daylight hours of high pH (Wahl et al., 2018; Young and Gobler, 2018). In contrast, recent experiments on larvae of different bivalve species found that simulated pH fluctuating of ~0.3 units around ambient and low pH conditions often did not mitigate the negative impacts of reduced pH on larval

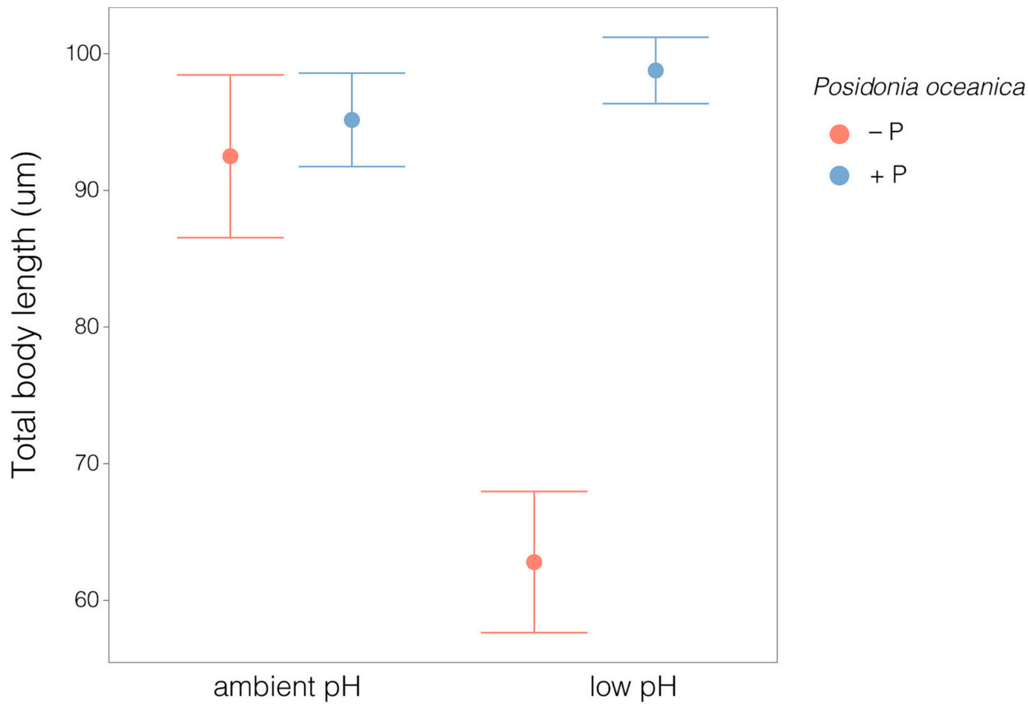


Fig. 4. Total body length (mean ± SE, $n = 5$) of echinopluteus larvae growing with or without *P. oceanica* plants under ambient and low pH conditions, for the first experimental run.

development and growth (Frieder et al., 2014; Kapsenberg et al., 2018), suggesting that the direction and magnitude of the effects of these variations can be species- and life-stage specific (Kapsenberg and Cyronak, 2019; Krause-Jensen et al., 2016).

Here, in line with our hypotheses, we found that *P. oceanica* plants increased mean daily pH values of ~0.1 and ~0.15 units at ambient and low pH treatments, respectively, compared to tanks without plants. These changes in pH and seawater chemistry likely provided a temporal refugia for sea urchin larvae exposed to low pH, as the percentage of larval abnormal development and their total body length significantly decreased and increased, respectively, in tanks with *P. oceanica* compared to those without plants. In particular, the total body length of echinoplutei growing at low pH within *P. oceanica* plants was comparable to those recorded at ambient pH. However, since the growth of echinoplutei was only assessed during the first experimental run, the experiment on the late larval development was not properly replicated. Although we allocated 5 tubes with several larvae for each tank, they were grouped within each tank exposed to different experimental conditions, generating problems of pseudoreplication. Further experiments are, thus, needed to provide a more robust assessment of the role of *P. oceanica* on the final development stages of sea urchin larvae exposed to low pH.

To the best of our knowledge, this is the first study experimentally investigating the effects of daily enhanced pH, generated biologically, on those stages of the sea urchin life-cycle that can be realistically influenced by the presence of benthic macrophytes. Although sea urchins can have long larval durations (i.e., ~30 days in *P. lividus,* García et al., 2015) and endure a wide range of environmental conditions during their planktonic life, the first and final stages of larval development are those more likely to be influenced by benthic communities. In particular, adult specimens of *P. lividus* are commonly found within *P. oceanica* meadows (Ceccherelli et al., 2009), which can also represent a suitable substrate for larval settlement (Privitera et al., 2011), having thus the potential to mitigate the negative effects of OA on the early and final stages of *P. lividus* larvae. The contrasting results between our work and those found in previous experiment with bivalve larvae (Clark and Gobler, 2016; Frieder et al., 2014; Kapsenberg et al., 2018) could be due to variations in OA sensitivity among early life stages of calcifying taxa. Alternatively, it could be the result of the different methods used to manipulate pH fluctuations as, in the aforementioned studies, pH changes were chemically (not biologically) induced (Clark and Gobler, 2016; Frieder et al., 2014; Kapsenberg et al., 2018) and may not be comparable to fluctuations of pH and associated carbonate chemistry imposed by biological processes. In addition, the presence of *P. oceanica* could potentially mitigate the negative effects of low pH on larval growth by increasing food availability (e.g., through enhanced dissolved organic materials; Barrón et al., 2014), other than pH. The role of resource availability in mediating the susceptibility of calcifying invertebrates, including sea urchins, to OA has been reported in previous studies (Foo et al., 2020; Pansch et al., 2014; Ramajo et al., 2016; Thomsen et al., 2013). For instance, Foo et al. (2020) found that low pH decreased and increased the arm length of sea urchin larvae growing in laboratory and at CO_2 vent site, respectively, and these divergent responses were probably driven by enhanced food availability in the field. However, under ambient pH conditions, echinoplutei generally display small arms when food is not limiting, as a strategy to allocate more resource to the growth of the juvenile rudiment and promote early settlement (Pedrotti and Fenaux, 1993; Soars et al., 2009). In our study, under ambient pH conditions, there were no differences in the total body length of echinoplutei grown with or without *P. oceanica* plants, suggesting that the positive response of larvae to increased pH caused by plant metabolism were not biased by differences in resource availability. Further experiments are, however, needed to understand the complex interplay between resource availability and pH variations in mediating larval development and growth under future OA scenario.

The role of vegetated habitats, such as seagrass meadows and macroalgal forests, in rescuing stress-sensitive species from increasingly adverse conditions due to climate change has recently garnered considerable attention to identify potential climate refugia and devised sound mitigation strategies, such as restoration or conservation plans (Bulleri et al., 2018; Falkenberg et al., 2021; Kapsenberg and Cyronak, 2019; Noisette et al., 2022). Here, we show that the presence of *P. oceanica* plants, by raising pH and associated carbonate saturation state through their metabolic activity, can buffer the negative effects of

OA on the calcifying larvae of the sea urchin, *P. lividus*.

Thus, the results of our study add to recent evidence that macrophyte-driven biogenic fluctuations can be managed as nature-based solution against climate-driven loss of biodiversity under future climates (Falkenberg et al., 2021; Noisette and Hurd, 2018; Wahl et al., 2018; Young and Gobler, 2018). However, it is worth noting that our study was performed under controlled laboratory conditions, thus neglecting the role of environmental factors in influencing the capacity and the extent of macrophyte metabolism to provide OA refugia. In particular, the existence of refugia seems to be primarily influenced by local hydrodynamic conditions, since high flow rates can override biologically driven pH variations at various spatial scales (from boundary layers to habitats) and neutralize the expected benefits of vegetated habitats (James et al., 2020; Noisette et al., 2022). Thus, taking into account variations of local hydrodynamic forces is necessary to improve our ability to predict the efficiency of chemical refugia in the face of climate changes.

Importantly, a benefactor species should be able to withstand adverse future conditions, without major changes in structural traits, such as morphology or density, that could undermine its ability to rescue other species (Bulleri et al., 2018). *P. oceanica*, like other macrophytes, is predicted to benefit, or to be unaffected by chemical changes in seawater driven by enhanced CO_2 concentrations (Cox et al., 2016; Hall-Spencer et al., 2008; Koch et al., 2013), having thus the ability to persist and deliver benefits to associated organisms under future OA scenario. Nonetheless, it is worth nothing that the ongoing anthropogenic CO_2 increase will be associated to increased thermal stress (Pachauri et al., 2014), potentially impairing macrophyte productivity and their ability to provide climate refugia (Collier et al., 2018). Hence, a thorough understanding on the impacts of multiple stressors on foundation species is critical to devise climate-ready solutions for assessing their potential to function as climate-rescuers for biodiversity under future climate scenarios.

Funding

This work was supported by the European Union's Horizon 2020 research and innovation program under grant agreement No 869300 FutureMARES.

CRediT authorship contribution statement

C. Ravaglioli: Conceptualization, Methodology, Investigation, Formal analysis, Writing – original draft. **L. De Marchi:** Conceptualization, Methodology, Investigation, Data curation, Writing – review & editing. **J. Giannessi:** Investigation, Data curation, Writing – review & editing. **C. Pretti:** Conceptualization, Methodology, Writing – review & editing. **F. Bulleri:** Conceptualization, Methodology, Supervision, Funding acquisition, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The datasets generated during the current study are available on Figshare.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.scitotenv.2023.167465.

References

Altieri, A.H., Silliman, B.R., Bertness, M.D., 2007. Hierarchical organization via a facilitation cascade in intertidal cordgrass bed communities. Am. Nat. 169, 195–206. https://doi.org/10.2307/4125317.

Barrón, C., Apostolaki, E.T., Duarte, C.M., 2014. Dissolved organic carbon fluxes by seagrass meadows and macroalgal beds. Front. Mar. Sci. 1.

Bulleri, F., Eriksson, B.K., Queirós, A., Airoldi, L., Arenas, F., Arvanitidis, C., Bouma, T.J., Crowe, T.P., Davoult, D., Guizien, K., 2018. Harnessing positive species interactions as a tool against climate-driven loss of coastal biodiversity. PLoS Biol. 16, e2006852.

Byrne, M., Hernández, J.C., 2020. Sea urchins in a high CO_2 world: impacts of climate warming and ocean acidification across life history stages. In: Developments in Aquaculture and Fisheries Science. Elsevier, pp. 281–297.

Byrne, M., Lamare, M., Winter, D., Dworjanyn, S.A., Uthicke, S., 2013. The stunting effect of a high CO2 ocean on calcification and development in sea urchin larvae, a synthesis from the tropics to the poles. Philos. Trans. R. Soc. B Biol. Sci. 368, 20120439.

Byrne, M., Foo, S.A., Ross, P.M., Putnam, H.M., 2020. Limitations of cross-and multigenerational plasticity for marine invertebrates faced with global climate change. Glob. Chang. Biol. 26, 80–102.

Ceccherelli, G., Pais, A., Pinna, S., Serra, S., Sechi, N., 2009. On the movement of the sea urchin *Paracentrotus lividus* towards *Posidonia oceanica* seagrass patches. J. Shellfish Res. 28, 397–403. https://doi.org/10.2983/035.028.0224.

Chan, K.Y.K., Grünbaum, D., O'Donnell, M.J., 2011. Effects of ocean-acidification-induced morphological changes on larval swimming and feeding. J. Exp. Biol. 214, 3857–3867. https://doi.org/10.1242/jeb.054809.

Clark, H.R., Gobler, C.J., 2016. Diurnal fluctuations in CO2 and dissolved oxygen concentrations do not provide a refuge from hypoxia and acidification for early-life-stage bivalves. Mar. Ecol. Prog. Ser. 558, 1–14.

Clements, J.C., Comeau, L.A., Carver, C.E., Mayrand, É., Plante, S., Mallet, A.L., 2018. Short-term exposure to elevated pCO_2 does not affect the valve gaping response of adult eastern oysters, Crassostrea virginica, to acute heat shock under an ad libitum feeding regime. J. Exp. Mar. Biol. Ecol. 506, 9–17. https://doi.org/10.1016/j.jembe.2018.05.005.

Collier, C.J., Langlois, L., Ow, Y., Johansson, C., Giammusso, M., Adams, M.P., O'Brien, K.R., Uthicke, S., 2018. Losing a winner: thermal stress and local pressures outweigh the positive effects of ocean acidification for tropical seagrasses. New Phytol. 219, 1005–1017. https://doi.org/10.1111/nph.15234.

Cornwall, C.E., Comeau, S., Kornder, N.A., Perry, C.T., van Hooidonk, R., DeCarlo, T.M., Pratchett, M.S., Anderson, K.D., Browne, N., Carpenter, R., 2021. Global declines in coral reef calcium carbonate production under ocean acidification and warming. Proc. Natl. Acad. Sci. 118, e2015265118 https://doi.org/10.1073/pnas.2015265118.

Cox, T.E., Gazeau, F., Alliouane, S., Hendriks, I.E., Mahacek, P., Le Fur, A., Gattuso, J.-P., 2016. Effects of in situ CO_2 enrichment on structural characteristics, photosynthesis, and growth of the Mediterranean seagrass Posidonia oceanica. Biogeosciences 13, 2179–2194. https://doi.org/10.5194/bg-13-2179-2016.

Dattolo, E., Ruocco, M., Brunet, C., Lorenti, M., Lauritano, C., D'Esposito, D., de Luca, P., Sanges, R., Mazzuca, S., Procaccini, G., 2014. Response of the seagrass *Posidonia oceanica* to different light environments: insights from a combined molecular and photo-physiological study. Mar. Environ. Res. 101, 225–236. https://doi.org/10.1016/j.marenvres.2014.07.010.

Dickson, A.G., Millero, F.J., 1987. A comparison of the equilibrium constants for the dissociation of carbonic acid in seawater media. Deep Sea Res. Part A. Oceanogr. Res. Pap. 34, 1733–1743. https://doi.org/10.1016/0198-0149(87)90021-5.

Doney, S.C., Fabry, V.J., Feely, R.A., Kleypas, J.A., 2009. Ocean acidification: the other CO_2 problem. Annu. Rev. Mar. Sci. 1, 169–192. https://doi.org/10.1146/annurev.marine.010908.163834.

Dworjanyn, S.A., Byrne, M., 2018. Impacts of ocean acidification on sea urchin growth across the juvenile to mature adult life-stage transition is mitigated by warming. Proc. R. Soc. B Biol. Sci. 285, 20172684.

Espinel-Velasco, N., Agüera, A., Lamare, M., 2020. Sea urchin larvae show resilience to ocean acidification at the time of settlement and metamorphosis. Mar. Environ. Res. 159, 104977. https://doi.org/10.1016/j.marenvres.2020.104977.

Falkenberg, L.J., Scanes, E., Ducker, J., Ross, P.M., 2021. Biotic habitats as refugia under ocean acidification. Conserv. Physiol. 9, coab077.

Foo, S.A., Koweek, D.A., Munari, M., Gambi, M.C., Byrne, M., Caldeira, K., 2020. Responses of sea urchin larvae to field and laboratory acidification. Sci. Total Environ. 723, 138003. https://doi.org/10.1016/j.scitotenv.2020.138003.

Frieder, C.A., Gonzalez, J.P., Bockmon, E.E., Navarro, M.O., Levin, L.A., 2014. Can variable pH and low oxygen moderate ocean acidification outcomes for mussel larvae? Glob. Chang. Biol. 20, 754–764.

García, E., Hernández, J.C., Clemente, S., Cohen-Rengifo, M., Hernández, C.A., Dupont, S., 2015. Robustness of Paracentrotus lividus larval and post-larval development to pH levels projected for the turn of the century. Mar. Biol. 162, 2047–2055. https://doi.org/10.1007/s00227-015-2731-8.

Gilman, S.E., Urban, M.C., Tewksbury, J., Gilchrist, G.W., Holt, R.D., 2010. A framework for community interactions under climate change. Trends Ecol. Evol. 25, 325–331. https://doi.org/10.1016/j.tree.2010.03.002.

Gribben, P.E., Byers, J.E., Clements, M., McKenzie, L.A., Steinberg, P.D., Wright, J.T., 2009. Behavioural interactions between ecosystem engineers control community species richness. Ecol. Lett. 12, 1127–1136. https://doi.org/10.1111/j.1461-0248.2009.01366.x.

Guettaf, M., San Martin, G.A., Francour, P., 2000. Interpopulation variability of the reproductive cycle of Paracentrotus lividus (Echinodermata: Echinoidea) in the south-western Mediterranean. J. Mar. Biol. Assoc. United Kingdom 80, 899–907. https://doi.org/10.1017/S0025315400002885.

Hall-Spencer, J.M., Rodolfo-Metalpa, R., Martin, S., Ransome, E., Fine, M., Turner, S.M., Rowley, S.J., Tedesco, D., Buia, M.-C., 2008. Volcanic carbon dioxide vents show ecosystem effects of ocean acidification. Nature 454, 96–99. https://doi.org/10.1038/nature07051.

Hendriks, I.E., Olsen, Y.S., Ramajo, L., Basso, L., Steckbauer, A., Moore, T.S., Howard, J., Duarte, C.M., 2014. Photosynthetic activity buffers ocean acidification in seagrass meadows. Biogeosciences 11, 333–346. https://doi.org/10.5194/bg-11-333-2014.

Hereu, B., Zabala, M., Linares, C., Sala, E., 2004. Temporal and spatial variability in settlement of the sea urchin *Paracentrotus lividus* in the NW Mediterranean. Mar. Biol. 144, 1011–1018. https://doi.org/10.1007/s00227-003-1266-6.

James, R.K., van Katwijk, M.M., van Tussenbroek, B.I., van der Heide, T., Dijkstra, H.A., van Westen, R.M., Pietrzak, J.D., Candy, A.S., Klees, R., Riva, R.E.M., Slobbe, C.D., Katsman, C.A., Herman, P.M.J., Bouma, T.J., 2020. Water motion and vegetation control the pH dynamics in seagrass-dominated bays. Limnol. Oceanogr. 65, 349–362. https://doi.org/10.1002/lno.11303.

Kapsenberg, L., Cyronak, T., 2019. Ocean acidification refugia in variable environments. Glob. Chang. Biol. 25, 3201–3214.

Kapsenberg, L., Miglioli, A., Bitter, M.C., Tambutté, E., Dumollard, R., Gattuso, J.-P., 2018. Ocean pH fluctuations affect mussel larvae at key developmental transitions. Proc. R. Soc. B 285, 20182381.

Karelitz, S., Lamare, M., Patel, F., Gemmell, N., Uthicke, S., 2020. Parental acclimation to future ocean conditions increases development rates but decreases survival in sea urchin larvae. Mar. Biol. 167, 1–16.

Koch, M., Bowes, G., Ross, C., Zhang, X., 2013. Climate change and ocean acidification effects on seagrasses and marine macroalgae. Glob. Chang. Biol. 19, 103–132. https://doi.org/10.1111/j.1365-2486.2012.02791.x.

Krause-Jensen, D., Marbà, N., Sanz-Martin, M., Hendriks, I.E., Thyrring, J., Carstensen, J., Sejr, M.K., Duarte, C.M., 2016. Long photoperiods sustain high pH in Arctic kelp forests. Sci. Adv. 2, e1501938.

Kroeker, K.J., Kordas, R.L., Crim, R.N., Singh, G.G., 2010. Meta-analysis reveals negative yet variable effects of ocean acidification on marine organisms. Ecol. Lett. 13, 1419–1434. https://doi.org/10.1111/j.1461-0248.2010.01518.x.

Kroeker, K.J., Kordas, R.L., Crim, R., Hendriks, I.E., Ramajo, L., Singh, G.S., Duarte, C.M., Gattuso, J., 2013. Impacts of ocean acidification on marine organisms: quantifying sensitivities and interaction with warming. Glob. Chang. Biol. 19, 1884–1896. https://doi.org/10.1111/gcb.12179.

Kroeker, K.J., Kordas, R.L., Harley, C.D.G., 2017. Embracing interactions in ocean acidification research: confronting multiple stressor scenarios and context dependence. Biol. Lett. 13, 20160802. https://doi.org/10.1098/rsbl.2016.0802.

Kurihara, H., 2008. Effects of CO_2-driven ocean acidification on the early developmental stages of invertebrates. Mar. Ecol. Prog. Ser. 373, 275–284.

Lamare, M.D., Liddy, M., Uthicke, S., 2016. In situ developmental responses of tropical sea urchin larvae to ocean acidification conditions at naturally elevated pCO_2 vent sites. Proc. R. Soc. B Biol. Sci. 283, 20161506.

Martin, S., Richier, S., Pedrotti, M.-L., Dupont, S., Castejon, C., Gerakis, Y., Kerros, M.-E., Oberhänsli, F., Teyssié, J.-L., Jeffree, R., Gattuso, J.-P., 2011. Early development and molecular plasticity in the Mediterranean sea urchin Paracentrotus lividus exposed to CO_2-driven acidification. J. Exp. Biol. 214, 1357–1368. https://doi.org/10.1242/jeb.051169.

Mehrbach, C., Culberson, C.H., Hawley, J.E., Pytkowicx, R.M., 1973. Measurement of the apparent dissociation constants of carbonic acid in seawater at atmospheric pressure. Limnol. Oceanogr. 18, 897–907. https://doi.org/10.4319/lo.1973.18.6.0897.

Migliaccio, O., Pinsino, A., Maffioli, E., Smith, A.M., Agnisola, C., Matranga, V., Nonnis, S., Tedeschi, G., Byrne, M., Gambi, M.C., 2019. Living in future ocean acidification, physiological adaptive responses of the immune system of sea urchins resident at a CO_2 vent system. Sci. Total Environ. 672, 938–950. https://doi.org/10.1016/j.scitotenv.2019.04.005.

Nagelkerken, I., Connell, S.D., 2015. Global alteration of ocean ecosystem functioning due to increasing human CO_2 emissions. Proc. Natl. Acad. Sci. 112, 13272–13277. https://doi.org/10.1073/pnas.1510856112.

Noisette, F., Hurd, C., 2018. Abiotic and biotic interactions in the diffusive boundary layer of kelp blades create a potential refuge from ocean acidification. Funct. Ecol. 32, 1329–1342. https://doi.org/10.1111/1365-2435.13067.

Noisette, F., Pansch, C., Wall, M., Wahl, M., Hurd, C.L., 2022. Role of hydrodynamics in shaping chemical habitats and modulating the responses of coastal benthic systems to ocean global change. Glob. Chang. Biol. 28, 3812–3829. https://doi.org/10.1111/gcb.16165.

Pacella, S.R., Brown, C.A., Waldbusser, G.G., Labiosa, R.G., Hales, B., 2018. Seagrass habitat metabolism increases short-term extremes and long-term offset of CO_2 under future ocean acidification. Proc. Natl. Acad. Sci. 115, 3870–3875.

Pachauri, R.K., Allen, M.R., Barros, V.R., Broome, J., Cramer, W., Christ, R., Church, J.A., Clarke, L., Dahe, Q., Dasgupta, P., 2014. Climate change 2014: synthesis report. In: Contribution of Working Groups I, II and III to the Fifth Assessment Report of the Intergovernmental Panel on Climate Change. Ipcc.

Pansch, C., Schaub, I., Havenhand, J., Wahl, M., 2014. Habitat traits and food availability determine the response of marine invertebrates to ocean acidification. Glob. Chang. Biol. 20, 765–777. https://doi.org/10.1111/gcb.12478.

Pedrotti, A.L., Fenaux, L., 1993. Effects of food diet on the survival, development and growth rates of two cultured echinoplutei (*Paracentrotus lividus* and *Arbacia lixula*). Invertebr. Reprod. Dev. 24, 59–69. https://doi.org/10.1080/07924259.1993.9672332.

Pfister, C.A., Altabet, M.A., Weigel, B.L., 2019. Kelp beds and their local effects on seawater chemistry, productivity, and microbial communities. Ecology 100, e02798.

Pinna, S., Pais, A., Campus, P., Sechi, N., Ceccherelli, G., 2012. Habitat preferences of the sea urchin *Paracentrotus lividus*. Mar. Ecol. Prog. Ser. 445, 173–180.

Prado, P., Romero, J., Alcoverro, T., 2009. Welcome mats? The role of seagrass meadow structure in controlling post-settlement survival in a keystone sea-urchin species. Estuar. Coast. Shelf Sci. 85, 472–478. https://doi.org/10.1016/j.ecss.2009.09.012.

Privitera, D., Noli, M., Falugi, C., Chiantore, M., 2011. Benthic assemblages and temperature effects on *Paracentrotus lividus* and *Arbacia lixula* larvae and settlement. J. Exp. Mar. Biol. Ecol. 407, 6–11. https://doi.org/10.1016/j.jembe.2011.06.030.

Ramajo, L., Pérez-León, E., Hendriks, I.E., Marbà, N., Krause-Jensen, D., Sejr, M.K., Blicher, M.E., Lagos, N.A., Olsen, Y.S., Duarte, C.M., 2016. Food supply confers calcifiers resistance to ocean acidification. Sci. Rep. 6, 19374. https://doi.org/10.1038/srep19374.

Riahi, K., Rao, S., Krey, V., Cho, C., Chirkov, V., Fischer, G., Kindermann, G., Nakicenovic, N., Rafaj, P., 2011. RCP 8.5—a scenario of comparatively high greenhouse gas emissions. Clim. Chang. 109, 33. https://doi.org/10.1007/s10584-011-0149-y.

Ricart, A.M., Gaylord, B., Hill, T.M., Sigwart, J.D., Shukla, P., Ward, M., Ninokawa, A., Sanford, E., 2021a. Seagrass-driven changes in carbonate chemistry enhance oyster shell growth. Oecologia 196, 565–576. https://doi.org/10.1007/s00442-021-04949-0.

Ricart, A.M., Ward, M., Hill, T.M., Sanford, E., Kroeker, K.J., Takeshita, Y., Merolla, S., Shukla, P., Ninokawa, A.T., Elsmore, K., Gaylord, B., 2021b. Coast-wide evidence of low pH amelioration by seagrass ecosystems. Glob. Chang. Biol. 27, 2580–2591. https://doi.org/10.1111/gcb.15594.

Ruocco, M., Entrambasaguas, L., Dattolo, E., Milito, A., Marín-Guirao, L., Procaccini, G., 2021. A king and vassals' tale: molecular signatures of clonal integration in Posidonia oceanica under chronic light shortage. J. Ecol. 109, 294–312. https://doi.org/10.1111/1365-2745.13479.

Sartori, D., Pellegrini, D., Macchia, S., Gaion, A., 2016. Can echinoculture be a feasible and effective activity? Analysis of fast reliable breeding conditions to promote gonadal growth and sexual maturation in *Paracentrotus lividus*. Aquaculture 451, 39–46. https://doi.org/10.1016/j.aquaculture.2015.08.037.

Sass, E., Ben-Yaakov, S., 1977. The carbonate system in hypersaline solutions: dead sea brines. Mar. Chem. 5, 183–199. https://doi.org/10.1016/0304-4203(77)90006-8.

Simon, T.N., Levitan, D.R., 2011. Measuring fertilization success of broadcast-spawning marine invertebrates within seagrass meadows. Biol. Bull. 220, 32–38.

Soars, N., Prowse, T., Byrne, M., 2009. Overview of phenotypic plasticity in echinoid larvae, "Echinopluteus transversus" type vs. typical echinoplutei. Mar. Ecol. Prog. Ser. 383, 113–125.

Steller, D., Cáceres-Martínez, C., 2009. Coralline algal rhodoliths enhance larval settlement and early growth of the Pacific calico scallop Argopecten ventricosus. Mar. Ecol. Prog. Ser. 396, 49–60.

Stumpp, M., Wren, J., Melzner, F., Thorndyke, M.C., Dupont, S.T., 2011. CO_2 induced seawater acidification impacts sea urchin larval development I: elevated metabolic rates decrease scope for growth and induce developmental delay. Comp. Biochem. Physiol. Part A Mol. Integr. Physiol. 160, 331–340. https://doi.org/10.1016/j.cbpa.2011.06.022.

Stumpp, M., Hu, M.Y., Melzner, F., Gutowska, M.A., Dorey, N., Himmerkus, N., Holtmann, W.C., Dupont, S.T., Thorndyke, M.C., Bleich, M., 2012. Acidified seawater impacts sea urchin larvae pH regulatory systems relevant for calcification. Proc. Natl. Acad. Sci. 109, 18192–18197. https://doi.org/10.1073/pnas.1209174109.

Stumpp, M., Hu, M., Casties, I., Saborowski, R., Bleich, M., Melzner, F., Dupont, S., 2013. Digestion in sea urchin larvae impaired under ocean acidification. Nat. Clim. Chang. 3, 1044–1049. https://doi.org/10.1038/nclimate2028.

Sunday, J.M., Fabricius, K.E., Kroeker, K.J., Anderson, K.M., Brown, N.E., Barry, J.P., Connell, S.D., Dupont, S., Gaylord, B., Hall-Spencer, J.M., 2017. Ocean acidification can mediate biodiversity shifts by changing biogenic habitat. Nat. Clim. Chang. 7, 81–85. https://doi.org/10.1038/NCLIMATE3161.

Thomsen, J., Casties, I., Pansch, C., Körtzinger, A., Melzner, F., 2013. Food availability outweighs ocean acidification effects in juvenile Mytilus edulis: laboratory and field experiments. Glob. Chang. Biol. 19, 1017–1027. https://doi.org/10.1111/gcb.12109.

Torres, O., Kwiatkowski, L., Sutton, A.J., Dorey, N., Orr, J.C., 2021. Characterizing mean and extreme diurnal variability of ocean CO_2 system variables across marine environments. Geophys. Res. Lett. 48, e2020GL090228 https://doi.org/10.1029/2020GL090228.

Uthicke, S., Ebert, T., Liddy, M., Johansson, C., Fabricius, K.E., Lamare, M., 2016. Echinometra sea urchins acclimatized to elevated pCO2 at volcanic vents outperform those under present-day pCO2 conditions. Glob. Chang. Biol. 22, 2451–2461.

Wahl, M., Schneider Covachã, S., Saderne, V., Hiebenthal, C., Müller, J.D., Pansch, C., Sawall, Y., 2018. Macroalgae may mitigate ocean acidification effects on mussel calcification by increasing pH and its fluctuations. Limnol. Oceanogr. 63, 3–21.

Young, C.S., Gobler, C.J., 2018. The ability of macroalgae to mitigate the negative effects of ocean acidification on four species of North Atlantic bivalve. Biogeosciences 15, 6167–6183.

Young, C.S., Sylvers, L.H., Tomasetti, S.J., Lundstrom, A., Schenone, C., Doall, M.H., Gobler, C.J., 2022. Kelp (Saccharina latissima) mitigates Coastal Ocean acidification and increases the growth of North Atlantic bivalves in lab experiments and on an oyster farm. Front. Mar. Sci. 513.